MinerRAG: A Trustworthy GeoAI System for Mining Stope Safety

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***Abstract*—Mining underground natural resources involves complex risks related to the environment, geology, and ore distribution. Although LLM-based GeoAI offers new opportunities, stope safety decision-making faces two major challenges: multi-source heterogeneous data makes it difficult to maintain stable yet experience-aware domain knowledge, and classic RAG cannot fully exploit correlations among spatiotemporal features, mining operations, and domain knowledge. To address these challenges, we propose MinerRAG, a trustworthy GeoAI system that enhances RAG with a Stope Knowledge Graph (SKG) for traceable consultative results. MinerRAG unifies multi-modal data, incrementally maintains the SKG through evidence-preserving factual relation edges and routed structural attachment edges, and improves reliability through spatiotemporal, specific-knowledge, and knowledge-topology correlations. Deployed in the Jinchuan Nickel and Cobalt Mine, MinerRAG manages more than 10,000 files with 1,156 entities and 1,184 relations. Knowledge update evaluation proves that the factual-edge precision over LLM-Direct improves from 22.76% to 26.47% with complete evidence, and the structural-attachment Top-1 accuracy improves from 68.29% to 78.05%. MinerRAG's demo video can be viewed at: https://youtu.be/mb9GdCTf5OY.**

***Index Terms*—knowledge graph, RAG, decision-making, mining stope safety, domain practice**

I. Introduction

Critical underground resources, such as nickel and cobalt, are widely used in key domains such as new energy, communications, and aerospace. Their mining faces immense safety factors in highly complex environments, including diverse surrounding rock properties, uneven ore body distribution, and extreme stress conditions. Therefore, decision support for mining stope safety is highly valuable in practice. Recently, the prosperity of Large Language Models (LLMs) presents GeoAI (AI for geology) and offers a promising avenue for intelligent decision-making.

However, AI usage in mining stope safety faces two critical limitations. (1) Due to the problems of multi-source heterogeneity and data quality, it is not trivial to maintain stable but experience-aware domain knowledge. In practice, safety-related data is distributed across business departments (e.g. equipment, mining, technology), including academic papers in PDF format, technical reports in DOC format, monthly dispatch presentations in PPT format, and engineering designs in CAD format. Such data is inconsistent and partial, while holistic domain knowledge such as geology, processes, and regulations should be up-to-date and empirical. (2) LLM-based GeoAI has inherent hallucinations, and domain-specific spatiotemporal correlations are implicitly scattered, making it difficult to guarantee trustworthy consultative results with traceable knowledge support. The design of 3D longitudinal stope inherently involves spatial relationships across panels and levels; operational procedures follow strict temporal sequences: excavation must precede backfilling. A given level of a panel must be in one of these states: planned excavation, ongoing excavation, planned backfill, ongoing backfill, and completed backfill, and different risks would be caused by environmental factors, such as rainfall or snowfall. Without delicate considerations, the widely adopted Retrieval-Augmented Generation (RAG), with inevitable hallucinations, cannot produce trustworthy consultative results.

To overcome such limitations, we propose **MinerRAG**, a trustworthy GeoAI system for mining stope safety. Our key contributions are summarized as follows.

- Through standardized pipelines, multi-modal data is reconstructed in unified representations, and the SKG is incrementally maintained, preserving factual evidence edges and structural attachment edges.
- For reliable consultative results, MinerRAG enhances RAG with traceable domain knowledge considering three hidden correlations: *Spatiotemporal Correlation* combines temporal event sequences with spatial physical topology of the mining stope; *Specific Knowledge Correlation* fuses high recall from domain knowledge with highly relevant supplements from general knowledge; *Knowledge Topology Correlation* emphasizes traceable results in domain knowledge even when ambiguous queries are described in natural language.
- MinerRAG is deployed in the Jinchuan Nickel and Cobalt Mine, and its practical evaluations quantify the trustworthiness gains.

II. Architecture and Key Modules

The MinerRAG architecture, shown in Figure 1, contains three key layers, which will be discussed in detail in the following sections.

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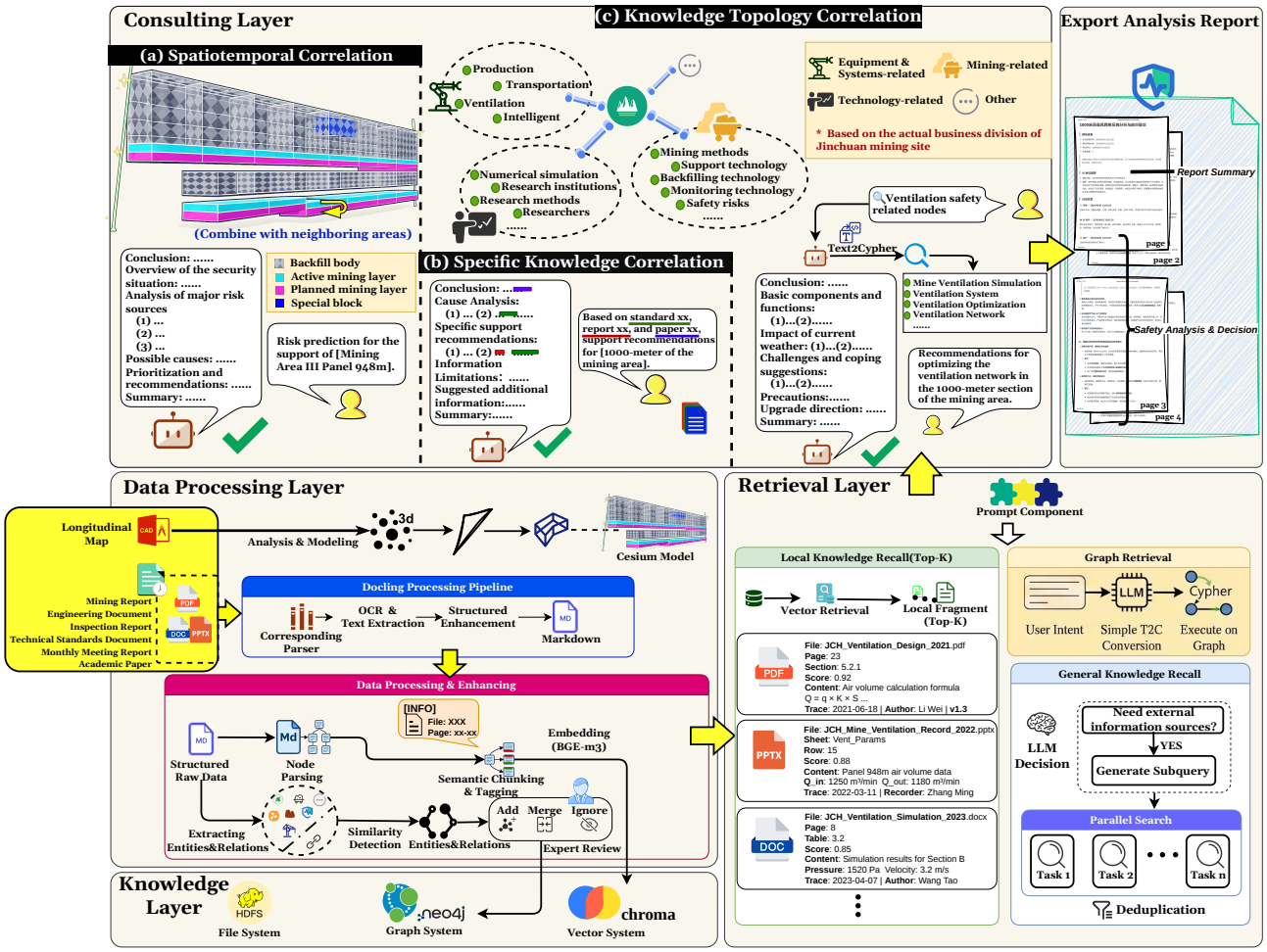


Fig. 1. MinerRAG Architecture

A. Data Processing Layer

The *Data Processing Layer* provides consistent data and domain knowledge for stope safety against multi-source heterogeneity and varying data quality.

In practice, domain knowledge for stope safety is distributed across multiple business departments, including mining, equipment, and technology management, and exists in diverse formats such as academic papers (PDF), technical reports (DOC), monthly dispatch presentations (PPT), and engineering design drawings (CAD). On such multi-modal data, a standardized pipeline on Docling procedure [1], [2] is designed to transform heterogeneous documents into a consistent unified representation. First, through document parsing, OCR-based text extraction, structure preservation, and semantic enhancement, unstructured and semi-structured contents are converted into lightweight structural Markdown format. We argue that the document's structure implies semantics. Therefore, heading hierarchy, table row/column, and paragraph order are kept here as much as possible. Subsequently, semantic chunking and metadata tagging are performed to preserve contextual integrity and traceability, while dense vector embeddings are generated and indexed for downstream retrieval. Based on the native heading tree, specific geological parameters or safety regulations can be recorded. Each chunk is then automatically injected with metadata labels, such as the source file name and specific page number, to facilitate filtering and traceability. Domain-specific *BGE-m3* embedding is adopted to map these chunks into dense high-dimensional vectors [3], which would be indexed in *Chroma* vector database. Finally, the unified data representation from pipeline bridges heterogeneous business data and machine-readable knowledge, and provides data foundation for *Retrieval Layer*.

Beyond unified representations, MinerRAG incrementally maintains a Stope Knowledge Graph (SKG) through a top-down and bottom-up mechanism. Domain experts first define stable categories and boundaries, such as mining operations, equipment systems, and technical management, forming a business-oriented skeleton. New knowledge can then be represented as two types of edges. The edge of factual relation preserves source segments, evidence text, and canonical relation to support provenance-based verification and canonicalization [4], [5]. The edge of structural attachment connects entities

to the SKG skeleton, where parent-node profiles and Jaro-Winkler similarity are fused into a routing score for automatic insertion, expert review, or temporary deferral. It follows the intuition of taxonomy expansion under domain constraints [6]. This procedure keeps knowledge updates traceable, structurally consistent, and empirically reliable.

B. Retrieval Layer

The *Retrieval Layer* serves as the central logic hub of MinerRAG to deliver consultative results by enhancing RAG with traceable domain knowledge. Building on the unified representations and the SKG, it coordinates local knowledge recall, graph recall, and selective general knowledge supplementation.

1) Local Retrieval: The first path, Local Retrieval, leverages the enterprise's proprietary data to guarantee high recall of domain-specific knowledge. This path employs two mechanisms: Vector Recall and Graph Recall.

In *Vector Recall*, queries are encoded into high-dimensional embeddings and matched against processed document chunks, while metadata labels trace each chunk back to its source document in pages [7].

In *Graph Recall*, natural language queries are translated into graph queries to traverse the SKG, identifying relevant entities, relations, and reasoning paths [8]. The retrieved evidence is fused in an evidence-first order: document chunks provide textual support, SKG paths provide entity-relation constraints, and candidate answers retain their provenance before generation.

2) Web Retrieval: The second path, Web Retrieval, supplements local knowledge with general knowledge to enhance contextual relevance, particularly for dynamic environmental factors or gaps in local evidence.

In the *General Knowledge Recall* process, MinerRAG evaluates the coverage and integrity of queries to determine whether external resources should be consulted. If local evidence is insufficient, queries are decomposed into sub-questions, and relevant information is retrieved from trusted external sources. The retrieved information is cleaned and added as supplements rather than replacing local evidence.

By dual retrieval paths, MinerRAG makes consultation results comprehensive, traceable, and aligned with the spatial, temporal, and operational context of mining stope, forming a reliable foundation for downstream decision-making.

C. Consulting Layer

The *Consulting Layer* synthesizes retrieved knowledge to provide trustworthy consultative results for mining stope safety. Three consultative entries are supported, and each entry is designed around a specific hidden correlation in mining safety scenarios. These three hidden correlations capture the spatial-temporal characteristics of mining operations, the complementary relationship between domain knowledge and general knowledge, and the structural dependencies embedded in the knowledge graph. Through these correlations, MinerRAG transforms retrieved evidence into traceable results for business users.

1) 3D Entry with Spatiotemporal Correlation: The 3D Entry binds a selected stope to panel-level topology and operational states. Because excavation, backfilling, and environmental disturbance follow strict temporal constraints [9], MinerRAG jointly analyzes neighboring stope, current operation states, and weather factors before generating a consultation.

2) Document Entry with Specific Knowledge Correlation: The Document Entry restricts consultation to given regulations, engineering reports, or geological investigations. MinerRAG prioritizes *Local Retrieval* within the specified scope and uses *Web Retrieval* only as a supplement when local evidence is insufficient, balancing compliance with evolving external knowledge.

3) Graph Entry with Knowledge Topology Correlation: The Graph Entry exposes the five-layer knowledge topology (mine area–category–domain–instance–keyword). Upon *Graph Recall*, MinerRAG maps query intentions to graphical paths to retrieve, and the subgraph as traceable result is used for a consultation. Since consultations are anchored to explicit entities, relations, and source evidences, topology-aware consultation improves credibility and interpretability.

III. Evaluation in a Practical Project

MinerRAG has been deployed in the Jinchuan Nickel and Cobalt Mine, one of China's largest underground mining enterprises. The system has imported over 10,000 files with multimodal data, including 8000+ academic papers, 3000+ technical reports, 200+ monthly dispatch presentations, and 1000+ engineering designs. On these data, MinerRAG maintains domain knowledge with 1,156 entities and 1,184 relations, and provides configurable consultative service through DeepSeek-V4-Pro, Qwen 3.5, and ChatGLM 4.7.

To quantify the trustworthiness of knowledge updates, MinerRAG's SKG update component, MinerRAG-KG, is evaluated on a manually annotated subset from Jinchuan No. 2 area [10]. The subset contains 14 papers, 728 text segments, 585 candidate factual relation segments, and 60 valid factual relation edges; for edges of structural attachment, 117 entities are split into 35 validation and 82 test samples, and Table I reports Top-1 accuracy on the test set. Baselines include Rule-Pattern, LLM-Direct, and NameSim respectively representing rule-based extraction, zero-shot LLM relation extraction, and shallow taxonomy-placement matching [6], [11]. For factual relation edges, source/evidence completeness is also a metric to indicate traceability.

TABLE I
BASELINE COMPARISON ON SKG UPDATES

Task	Method	Metric	Score	Src./Evid.
Factual edge	Rule-Pattern	Precision	8.76%	100%/100%
Factual edge	LLM-Direct	Precision	22.76%	0%/0%
Factual edge	MinerRAG-KG	Precision	**26.47%**	100%/100%
Structural attach.	NameSim	Top-1 Acc.	36.59%	–
Structural attach.	LLM-Direct	Top-1 Acc.	68.29%	–
Structural attach.	MinerRAG-KG	Top-1 Acc.	**78.05%**	–

As shown in Table I, MinerRAG-KG improves the factual-edge precision over LLM-Direct while retaining complete source and evidence, making candidate relations auditable rather than opaque triples. The comparison is auditability-oriented: the full KG update evaluation shows that LLM-Direct attains higher recall (46.67% vs. 30.00%), whereas MinerRAG-KG deliberately filters low-quality candidates before knowledge insertion. For structural attachment, MinerRAG-KG improves Top-1 accuracy from 68.29% to 78.05% over LLM-Direct and from 36.59% to 78.05% over NameSim. The Jaro-Winkler routing signal is selected on the validation set, with $\alpha = 0.50$, $T_a = 0.80$, and $T_r = 0.75$. Using this signal, automatic structural attachment reaches 86.96% precision and intercepts 83.33% of incorrect attachments before graph insertion.

The case study of Jinchuan No. 2 area further shows how trustworthy updates separate candidate generation from final graph insertion. One newly added standard document yields 19 candidate knowledge items, and after routing and manual verification, the updated results involve 7 newly created nodes, 19 merged nodes, 13 newly added relations, and 3 discarded relations. In daily operation, MinerRAG also supports exporting archival reports with traceable citation sources. Consultative responses can be selected and composed into a standardized PDF report with title, summary, and logically connected sections.

These results clarify two design motivations. The first is to preserve traceable factual knowledge updates. Rule-Pattern preserves source and evidence because it is tied to trigger windows, but its precision is limited by the complex semantic expressions in stope-domain text. In contrast, LLM-Direct improves semantic extraction but outputs opaque triples without source/evidence traceability. The second is to ensure structurally consistent taxonomy expansion. Likewise, name-only entity matching is too weak for fine-grained stope taxonomy, because it ignores the surrounding SKG skeleton and the semantic role of the parent node. MinerRAG-KG sits between these extremes: it keeps source segments, evidence text, and canonical relation types for factual relation edges, and it uses parent-node profiles plus routing thresholds for structural attachment edges.

The practical implication is a clear separation between knowledge generation and insertion. Candidate knowledge is generated first, then verified, and only the trustworthy subset is written into the SKG. This design keeps updates traceable while linking user-facing consultations and archival reports to the same evidence chain.

IV. Conclusion

To address the challenges of maintaining stable yet experience-aware domain knowledge from heterogeneous mining data and generating reliable consultative results with traceable knowledge, we present MinerRAG, a trustworthy GeoAI system for mining stope safety. MinerRAG maintains multi-modal knowledge with an enhanced RAG framework, incrementally maintains the SKG through evidence-preserving and routed graph updates, and improves consultation reliability considering three types of correlations. Our work has been successfully deployed in the Jinchuan Nickel and Cobalt Mine, supports daily consultation for stope safety, and demonstrates a trustworthy solution in an industrial scenario.

The current evaluation mainly quantifies the trustworthiness of SKG updates and reports industrial deployment evidence. Future work will extend trustworthy evaluation to enable comprehensive end-to-end comparisons of consultative answer quality across diverse LLM backbones, user roles, and risk scenarios. We will also explore the integration of sensory data, agent-based workflows, and time-series foundation models for more proactive and intelligent early-warning in mining operations.

Acknowledgment

This work was supported by the Key Project of Graduate Education and Teaching Reform of North China University of Technology.

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